



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

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1. CHEMICAL IDENTIFICATION

Product Identification

Product name: carbosulfan 20% W/V EC

Product use: Insecticide

Relevant identified uses of the substance/mixture and uses advised against:

Used for the control of a wide range of soil-dwelling and foliar insect pests. Examples of uses include control of millipedes, springtails, symphylids, wireworms, pygmy mangold beetles, frit flies, white grubs, aphids, caterpillars, flea beetles, Colorado beetles, stem borers, leafhoppers, plant hoppers, codling moth, scales and free-living nematodes. The product is used in a wide range of crops, e.g. cotton, sugar beet, potatoes, rice, top fruit, citrus, maize, vegetables, sugar cane and coffee.

2. Hazard Identification

Adverse Human Health Effects

Effects from overexposure may result from either swallowing, inhaling or coming into contact with the skin or eyes. Conditions of increased temperature and humidity facilitate skin absorption of this product, and therefore, promotes increased toxicity.

Environmental effects:

Carbosulfan is rapidly degraded in neutral soil (half-life < 5 days) with the degradation rate increasing as pH decreases. A major breakdown product is carbofuran, which degrades more slowly (half-life approx. 50 days). Carbosulfan is hydrolytically unstable in acid, with stability increasing with increasing pH. The bio-concentration potential of carbosulfan is low with log Pow of 3.3, and a measured BCF in fish of 990. Carbosulfan and its major breakdown product are unlikely to leach into groundwater.



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Physical and chemical hazard

Specific hazards:

Main symptoms: Symptoms of overexposure include headache, light-headedness, weakness, abdominal cramps, nausea, excessive salivation, perspiration, blurred vision, tearing, pin-point pupils, blue skin color, convulsions, tremor and coma.

Classification of product:

Emulsifiable Concentrate

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: 2,3-dihydro-2,2-dimethylbenzofuran-7-yl(dibutylaminothio)methylcarbamate
[IUPAC]

Common name: carbosulfan

CAS Number: 55285-14-8

Chemical family: Carbamate

Components contributing to the hazard or impurities:

Bis[(2,3-dihydro-2,2-dimethyl-7-benzofranyl)methylcarbamate]sulfide;

Bis(di-n-butylamino)trisulfide;

Carbofuran;

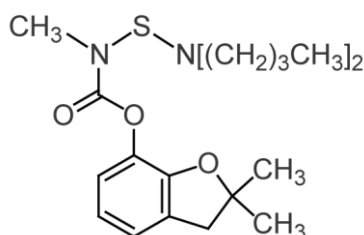
Bis(di-n-butylamino)tetrasulfide;

Bis(di-n-butylamino)pentasulfide

Organic solvent, Emulsifier, etc.

Formula: $C_{20}H_{32}N_2O_3S$

Molecular weight: 380.55 g/mol



4. FIRST AID MEASURES

Emergency overview

First aid measures:

Single ingestion,

Rinse mouth with water. Dilute by giving 1 or 2 glasses of water. Do not induce vomiting.

Never give anything by mouth to an unconscious person. See a medical doctor immediately.



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Short-term Eye contact,

Flush with water for at least 15 minutes. If irritation occurs and persists, contact a medical doctor.

Short-term Skin contact,

Remove contaminated clothing and thoroughly wash with soap and water. If irritation occurs and persists, contact a medical doctor.

Short-term Inhalation,

Remove to fresh air. If breathing difficulty or discomfort occurs and persists, see a medical doctor. If breathing has stopped, give artificial respiration and see a medical doctor immediately.

Important symptoms (detail of symptoms and effects should be given under heading 11)

Protection of first-aiders:

Notes to a physician: This product is moderately toxic if swallowed or inhaled, and slightly toxic to absorb through the skin. It is moderately irritating to skin and eyes. Carbosulfan is a reversible cholinesterase inhibitor. Atropine sulfate is antidotal. Support respiration as needed with removal of secretions, maintenance of a patent airway and, if necessary, artificial ventilation. If cyanosis is absent: Adults – start treatment by giving 2 mg atropine intravenously or intramuscularly, if necessary, and repeat with 0.4 – 2.0 mg atropine at 15 minute intervals until atropinization occurs (tachycardia, flushed skin, dry mouth, mydriasis); Children under 12 – initial dose = 0.05 mg/kg body weight and repeat dose = 0.02 – 0.05 mg/kg body weight. Use of oximes such as 2-PAM is controversial. Observe patient to insure that these symptoms do not recur as atropinization wears off. If in eyes, instill one drop of homatropine. Contains aromatic hydrocarbons that may produce a severe pneumonitis if aspirated during vomiting. Consideration should be given to gastric lavage with an endotracheal tube in place. Treatment is otherwise controlled removal of exposure followed by symptomatic and supportive care.

5. FIRE FIGHTING MEASURES

Extinguishing Media:

Foam, CO₂ or dry chemical. Soft stream water fog only if necessary. Contain all runoff.

Specific Hazards Arising from the Chemical:

Thermal decomposition can lead to release of irritating and toxic gases and vapors Combustible liquid.

Sensitivity to Mechanical Impact:

Not sensitive.

Sensitivity to Static Discharge:

Yes



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Fire Fighting Equipment and Precautions:

Isolate fire area. Evaluate downwind. Wear self-contained breathing apparatus and protective suit.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal Precautions:

Isolate and post spill area. Remove all sources of ignition. Wear suitable protective clothing, gloves and eye/face protection. For personal protection see section 8.

Other:

For further clean-up instructions, call FMC Emergency Hotline number listed in Section 1 "Product and Company Identification" above.

Environmental Precautions:

Keep people and animals away from and upwind of spill/leak. Keep material out of lakes, streams, ponds, and sewer drains.

Methods for Containment:

Dike to prevent runoff.

Methods for cleaning up:

Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. Clean and neutralize spill area, tools and equipment for a minimum contact of one hour with vinegar-alcohol solution, then by bleach soap and water. Absorb rinsate and add to the collected waste. Waste must be classified and labeled prior to recycling or disposal. Dispose of waste as indicated in Section 13.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Safe Handling:

Do not contaminate other pesticides, fertilizers, water, food, or feed by storage or disposal.

Storage Conditions:

Keep in a dry, cool and well-ventilated place. Keep away from open flames, hot surfaces and sources of ignition. Keep out of reach of children and animals. Keep/store only in original container

Incompatible products:

No information available Acids



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8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering measures to reduce exposure

Apply technical measures to comply with the occupational exposure limits. When working in confined spaces (tanks, containers, etc.), ensure that there is a supply of air suitable for breathing and wear the recommended equipment.

Personal Protective Equipment

Eye/Face Protection:

For dust, splash, mist or spray exposure, wear chemical protective goggles.

Skin/Body Protection:

Wear long-sleeved shirt, long pants, socks, shoes, chemical-resistant gloves and headgear.

Respiratory Protection:

For dust, splash, mist or spray exposures wear a full-face air-supplying respirator which is approved for pesticides (U.S. NIOSH/MSHA, EU CEN or comparable certification organization). Respirator use and selection must be based on airborne concentrations.

Hand Protection:

Wear protective gloves. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion and the contact time.

Hygiene measures:

Clean water should be available for washing in case of eye or skin contamination. Wash skin prior to eating, drinking, chewing gum or using tobacco. Shower or bathe at the end of working. Remove and wash contaminated clothing before re-use. Launder work clothing separately from regular household laundry.

General information:

If the product is used in mixtures, it is recommended that you contact the appropriate protective equipment suppliers. These recommendations apply to the product as supplied.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Liquid
Color:	Brown
Odor:	Aromatic hydrocarbon-like
Decomposition temperature:	> 146 °C
Explosion properties & Flash point:	> 56 °C
Vapor pressure:	0.12 mPa at 20 °C



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Vapor density:	0.98 g/ml at 20°C
Solubility in water:	With any scale soluble in water and most organic solvents.
Octanol/water partition coefficient:	Not available

10. STABILITY AND REACTIVITY

Conditions to avoid:

Excessive heat. Extreme risk of explosion by shock, friction, fire or other sources of ignition.

Material to avoid:

Mixing with strongly alkaline or acid substances should be avoided.

Chemical Stability:

Stable under recommended storage conditions

Possibility of Hazardous Reactions:

Contact with aqueous acids may produce carbofuran.

Hazardous polymerization:

Hazardous polymerization does not occur.

Hazardous composition products:

Burning produces obnoxious and toxic fumes: Carbon oxides (COx), Sulfur oxides, Nitrogen oxides (NOx).

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Carbosulfan:

LD50 (Oral, rat):	> 600 mg/kg
LD50 (Dermal, rat):	> 600 mg/kg
LC50 (Inhalation, rat, 4h):	> 6.5 mg/L
LC50 (Skin irritation, rat):	> 3,000 mg/kg
Eye irritation (rabbit):	Slight eye irritation in rabbits
Skin sensitization:	Not sensitizer

Reproductive/Teratology data

Long term toxicity, NOEL (2 years, rat): 35 mg/kg diet

Ingestion

ADI (JMPR): 0.03 mg/kg body weight

12. ECOLOGICAL INFORMATION

Mobility

Slightly mobile; not expected to reach groundwater.



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Persistence and degradability

Non-persistent. Readily hydrolyzed. Not readily biodegradable.

Bioaccumulation

The substance has a potential for bioconcentration.

Ecotoxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, Carbosulfan:

Fish – Acute toxicity:

LC50 (96 h): 0.015 mg/L

NOEC (21 d): 3.0 ug/L

Crustacea – Acute toxicity:

EC50 (48 h): 1.5 ug/L

NOEC (21 d): 3.0 ug/L

Algae – Acute toxicity:

EC50 (72 h): 47 mg/L

13. DISPOSAL CONSIDERATIONS

Waste disposal methods:

Improper disposal of excess pesticide, spray mixture, or rinsate is prohibited. If these wastes cannot be disposed of by use according to label instructions, contact appropriate disposal authorities for guidance.

Contaminated Packaging:

Containers must be disposed of in accordance with local, state and federal regulations. Refer to the product label for container disposal instructions.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No: 2991

Dangerous Goods Class: 9

Hazard Labeling and classification: II

Packaging Group: III

Marine pollutant: Yes

Description: UN2991, Carbamate pesticides, liquid, toxic, flammable (aromatic hydrocarbons, carbosulfan), 6.1 (3), PGIII, Marine Pollutant



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15. REGULATORY INFORMATION

Risk Phrases:

R23/25: Toxic by inhalation and if swallowed.

R25: Toxic if swallowed.

R26: Very toxic by inhalation.

R43: May cause sensitization by skin contact.

R50/53: Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Safety Phrases:

S24: Avoid contact with skin.

S28: After contact with skin, wash immediately with plenty of water and soap.

S36/37: Wear suitable protective clothing and gloves.

S37: Wear suitable gloves.

S38: In case of insufficient ventilation, wear suitable respiratory equipment.

S45: In case of accident or if you feel unwell, seek medical advice immediately (show the label where possible).

S60: This material and its container must be disposed of as hazardous waste.

S61: Avoid release to the environment. Refer to special instructions/safety data sheets.

S63: In case of accident by inhalation: remove casualty to fresh air and keep at rest.

16. OTHER INFORMATION

THEP WATANA CO., LTD. believes that the information and recommendations contained herein (including data and statements) are accurate as of the date hereof. NO WARRANTY OF FITNESS FOR ANY PARTICULAR PURPOSE, WARRANTY OF MERCHANTABILITY OR ANY OTHER WARRANTY, EXPRESSED OR IMPLIED, IS MADE CONCERNING THE INFORMATION PROVIDED HEREIN. The information provided herein relates only to the specified product designated and may not be applicable where such product is used in combination with any other materials or in any process. Use of this product is regulated by the U.S. Environmental Protection Agency (EPA). It is a violation of Federal law to use this product in a manner inconsistent with its labeling. Further, since the conditions and methods of use are beyond the control of THEP WATANA CO., LTD., THEP WATANA CO., LTD. expressly disclaims any and all liability as to any results obtained or arising from any use of the products or reliance on such information.